Alex Wang

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Education

University of Waterloo Sep 2023 – Apr 2028

Bachelor of Computer Science, AI Specialization

- Courses: Data Structures and Algorithms, Operating Systems, Computer Networks, Databases, Intro to ML

Experience

Geotab Sep 2026 – Dec 2026

Software Engineer Intern

- Built three Python benchmarks for a pre-launch API on 19M PostgreSQL rows; used **EXPLAIN ANALYZE** to trace **20 s** searches to full-table scans and validated an index cutting them to **35 ms**.
- Fixed two **permission bugs** in the Driver Coaching API: one exposed HR-sensitive data to unauthorized users, one let restricted users silently delete links they could not see; added **per-user checks** and restricted-user tests.
- Added driver filtering to the Safety Coaching API in **C#/.NET**, capping per-driver queries at **20 concurrent** to bound load on the data service.
- Fixed **error masking** in a shared **UnitOfWork** transaction library, where rollback failures overwrote the original exception and caused expected rejections to be misreported as errors.

iHuman Inc. Sep 2025 – Dec 2025

Software Developer Intern

- Built a **React/FastAPI** platform to automate multi-turn evaluation of English-conversation agents, cutting test rounds from **hours to minutes**; integrated Dify APIs with configurable personas and test suites.
- Decoupled conversation generation from evaluation via **asynchronous task orchestration** with bounded concurrency, scaling to **100+ dialogue turns per task** with live progress tracking and persisted results.
- Designed a hybrid scoring pipeline pairing rule-based metrics (vocabulary diversity, sentence length) with **LLM-as-judge** ratings of naturalness, correctness, and creativity to compare **10+ agent configurations**.

Joyson Robot Jan 2025 – Apr 2025

Software Developer Intern

- Cut redundant API calls by **80%** with a 6-hour refresh cache in a Python news aggregation service processing 100+ articles daily via third-party APIs, SQL filtering, and local **LLM summarization**.
- Fine-tuned a **vision-language model** with **LoRA** via LLaMA-Factory on **1,000+ annotated images** for fruit recognition and calorie-aware responses; built the pipeline to clean and convert annotations into training data.

Projects

Polyphony: Multi-Agent Deliberation Engine | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, Redis, MCP*

- Designed a **coordinator-free** engine where **11 concurrent LLM agents** share one event stream and each decides whether to speak; weighted votes run as **checkpointed LangGraph** workflows that resume after crashes.
- Enforced one global event order with **PostgreSQL** sequences so concurrent agents never read out of order, with **Redis** as the hot window and **pgvector** as long-term agent memory.
- Gated speech behind a cheap listen pass, so only **16% of 744** checks escalated to full generation; swapping the decider from Haiku to **Jev** cut its p95 latency from **4.1 s to under 0.5 s**.
- Exposed the engine as an **MCP** server, so external AI clients can ask the agents a question and get their decision.

Distributed Order Processing System | *Go, gRPC, Kafka, PostgreSQL, Redis, Docker, Kubernetes*

- Built a checkout system of **5 Go microservices** (gRPC, database per service) on **Kubernetes**; a **Saga** with guarded state transitions drives each order, so retries and crashes can never move it twice.
- Published compensation events through a **transactional outbox** to **Kafka** with **idempotent** consumers, so a crash mid-step loses no event and applies none twice.
- Treated payment timeouts as unknown outcomes: orders wait in a pending state while a recovery worker retries idempotent charges; **zero double charges** with **30%** of payments timing out.
- Prevented overselling with ordered row locks behind a DB **CHECK**, plus a **Redis Lua** gate for flash sales; **zero oversold** across 1,000 concurrent orders and 3,000 flash-sale buyers on 100 units.

Technical Skills

Languages: Python, C#, Go, SQL, TypeScript, JavaScript, Java, C/C++, Rust
Frameworks: .NET, FastAPI, Gin, gRPC, React, Spring Boot, Node.js, LangGraph, LLaMA-Factory
Tools: PostgreSQL, Redis, Kafka, MongoDB, Docker, Linux, GitLab CI/CD, AWS, Kubernetes, MCP, LLM Evals